Rot-KDEQ: Rotation and Density-Aware Quantization for Ultra-Low-Bit Post-Training Compression of Large Language Models

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Abstract

Post-training quantization (PTQ) is a key technique for enabling efficient inference of large language models on resource-constrained hardware. However, when pushed to ultra-low-bit precision, existing PTQ methods often suffer from severe performance degradation, limiting their practicality in real-world deployment. This challenge stems from the mismatch between quantization mechanisms and the complex statistical structure of activation distributions in modern LLMs.In this work, we propose Rot-KDEQ, a distribution-aware PTQ framework designed to operate reliably under ultra-low-bit settings. The core idea of Rot-KDEQ is to jointly control quantization error from complementary perspectives: global geometric reorganization and local density-aware range selection. By combining orthogonal rotation with per-channel statistical modeling, Rot-KDEQ constructs quantization intervals based on the high-density region of the activation distribution while separately handling extreme values.Extensive experiments across multiple LLMs families demonstrate that Rot-KDEQ consistently achieves superior robustness and accuracy under aggressive quantization, providing a practical solution for ultra-low-bit LLMs inference.

Keywords: Post-Training Quantization, Random Orthogonal Rotation, Per-Channel Statistical Modeling

1. Introduction

In recent years, Large Language Models (LLMs) have demonstrated remarkable progress in language understanding, text generation, and complex reasoning tasks. However, the rapid growth in model scale has resulted in prohibitive inference cost and memory footprint, severely hindering the deployment of LLMs on edge devices and resource-constrained hardware platforms [1, 2]. Compared with quantization-aware training (QAT) [3–5], which requires additional training overhead and access to training data, post-training quantization (PTQ) [6, 7] performs quantization without retraining, making it more flexible for deployment and widely adopted for compressing LLMs [8].Nevertheless, when quantization precision is reduced to 4-bit or lower, model performance often degrades drastically, significantly limiting the applicability of PTQ in practical LLMs inference systems [9–11].

The fundamental challenge of ultra-low-bit PTQ lies not merely in reduced numerical precision, but in the mismatch between quantization mechanisms and the statistical structure of activation distributions. Extensive studies have shown that activations in LLMs typically exhibit pronounced long-tailed characteristics, where the majority of values concentrate in a narrow high-density region, while a small number of extreme values reside in sparse tails [12, 13]. When quantization ranges are determined by global extrema statistics, these rare outliers disproportionately dominate scale selection, leading to excessively wide quantization intervals and amplified errors in the dense core region [14, 15]. This issue is further complicated by strong layer-wise and channel-wise heterogeneity in activation distributions, making it difficult for a single global scaling strategy to robustly accommodate the entire model, especially when activation and KV cache are quantized jointly [16–18].

To alleviate the adverse effects of long-tailed activation distributions, recent PTQ methods introduce geometric transformations to rebalance feature space representations [6, 19, 20], or employ heuristic clipping and thresholding strategies to suppress the influence of extreme values [21, 22]. While these approaches can partially mitigate distribution skewness, they suffer from inherent limitations. Methods that emphasize global geometric reorganization often fail to capture channel-wise density variations, which can still result in quantization range mismatch under ultra-low-bit settings. Conversely, approaches based on fixed or empirical range selection lack explicit modeling of the underlying distribution shape, making them unstable when confronted with complex, non-Gaussian activation patterns that vary significantly across layers. As a result, existing methods struggle to simultaneously control distribution imbalance at both global and local scales.

In this work, we propose **Rot-KDEQ**, a unified post-training quantization framework tailored for ultra-low-bit LLMs inference. The key idea of Rot-KDEQ is to jointly control quantization error from complementary perspectives: global geometric reorganization and local density-aware range selection. Specifically, random orthogonal rotation is employed to redistribute activation energy across dimensions under the principle of computational invariance [23], reducing the dominance of extreme values in individual channels. Meanwhile, per-channel Kernel Density Estimation (KDE) provides a non-parametric

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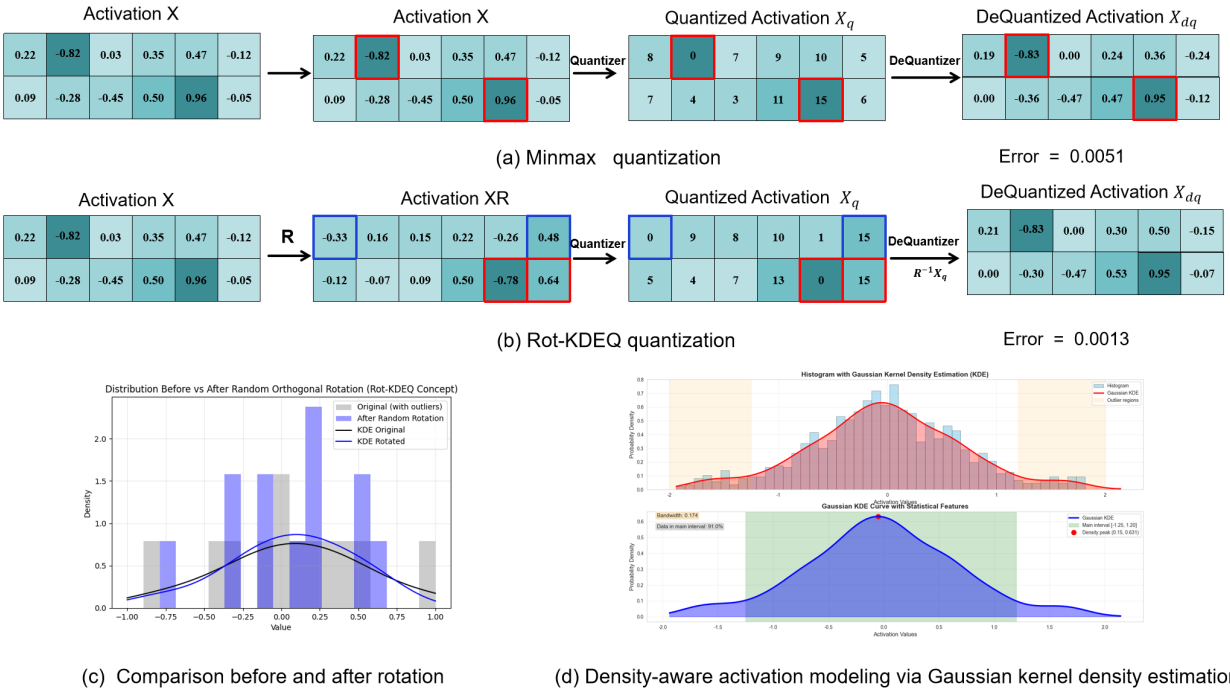


Figure 1: (a–b) Visualization of different quantization schemes, including min–max quantization and Rot-KDEQ. Color intensity indicates the value range of activation.(c) Effect of random orthogonal rotation in Rot-KDEQ. The original distribution (gray) exhibits a long-tailed shape with prominent outliers, whereas the rotated distribution (blue) becomes more symmetric and Gaussian like. The rotation disperses outlier energy across dimensions, reducing the influence of extreme activation outliers on quantization intervals.(d) Identification of the high-density core region of the activation distribution via Gaussian Kernel Density Estimation(KDE), with low-density tail regions treated as outliers. Determining quantization ranges based on this density-aware interval, rather than global extrema, significantly improves the robustness of scale and zero-point estimation in ultra-low-bit Post-Training Quantization(PTQ).

characterization of local activation density, enabling statistical identification of high-density core regions and low-density tails. Based on these estimates, Rot-KDEQ constructs adaptive quantization ranges and hybrid quantization factors that prioritize precision in the dominant activation mass while preserving information carried by extreme values. By integrating geometric rebalancing with density-aware modeling, Rot-KDEQ maintains stable and accurate quantization performance even under aggressive compression at 4-bit precision and below.

Model	LLaMA2-7B(↓)		LLaMA3-8B(↓)	
Bits	W4A4KV4	W3A3KV3	W4A4KV4	W3A3KV3
MinMax [22]	6.5	372.8	7.0	112.4
MSE [24]	6.9	276.1	7.2	165.7
Percentile [25]	6.3	12.5	7.8	14.5
KDE(Ours)	**5.9**	**11.1**	**6.9**	**12.3**

Table 1: **The rationality of quantitative parameters is a key prerequisite for the quantitative deployment of the model.** This figure compares the performance of the model with our method Kernel Density Estimation(KDE) under three standardized methods.

2. Background and Related Work

2.1. Post-Training Quantization for Large Language Models

Post-Training Quantization(PTQ) has become a mainstream solution for reducing the inference cost of LLMs [26–31]. Early works primarily focus on minimizing weight quantization error. For example, GPTQ [22] leverages second-order Hessian information to perform layer-wise reconstruction, significantly improving the stability of low-bit weight quantization. OmniQuant [32] jointly models weight and activation using calibration data to further enhance reconstruction accuracy. SmoothQuant [14] proposes an operator-level scale migration strategy to balances the dynamic ranges of activation and weight, effectively alleviating the difficulty of activation quantization. AWQ [10] identifies activation-sensitive weight channels and selectively preserves them under low-bit settings. QuIP [33] emploits compressed projections and basis transformations to reencode weight structures, exhibiting strong representational capacity for low-bit weight quantization.Despite their effectiveness in stabilizing weight quantization, these approaches often experience noticeable performance degradation when activations and KV cache are further compressed to 4-bit or lower precision, highlighting the growing importance of robust activation aware quantization strategies in ultra-low-bit regimes.

2.2. Outlier Suppression and Distribution Modeling

A large body of work has investigated strategies for mitigating the adverse effects of long-tailed activation distributions. SmoothQuant [14] reduces dynamic range imbalance by shifting activation scales into weight. ResQ [6] and QDrop [34] introduce channel-wise heterogeneity through mixed-precision quantization and stochastic dropping, respectively, to protect

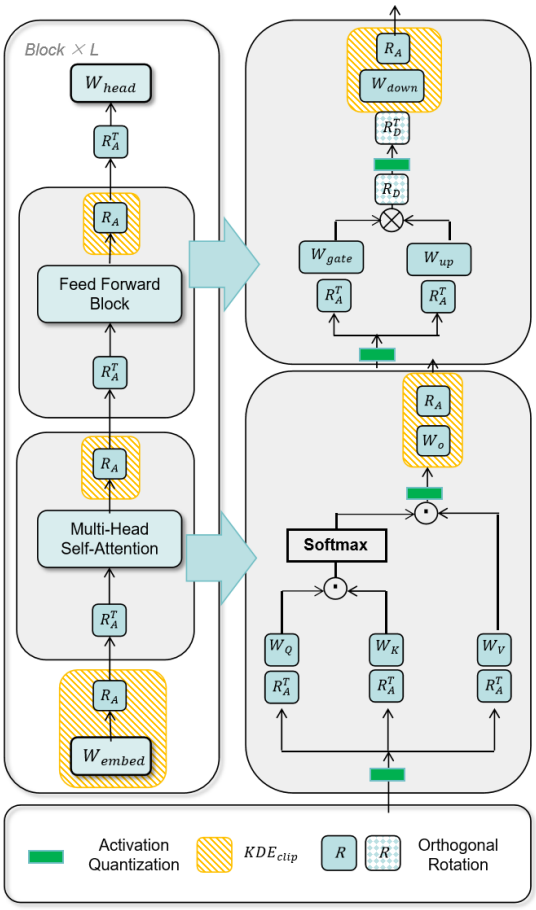


Figure 2: **Rot-KDEQ overview.**(A) Block-wise orthogonal rotation are fused into linear-layer weights to obtain isotropic activations with zero inference overhead; non-fusible FFN rotation are efficiently implemented via Hadamard transforms.(B) Per-channel Kernel Density Estimation(KDE) identifies high-density activation regions to construct adaptive quantization ranges, jointly suppressing outliers and preserving core precision under ultra-low-bit quantization.

sensitive regions containing extreme outliers. In addition, several methods rely on Empirical strategies, such as clipping or threshold adjustment,to suppress the excessive influence of outliers on quantization scales.While these methods can partially reduce the dominance of extreme activation values, they typically rely on global statistical assumptions. When activation distributions exhibit strong skewness or asymmetric multi-modal structures, heuristic or mean-based statistics often fail to robustly determine favorable quantization intervals, making it difficult to maintain control lable quantization error under ultra-low-bit constraints.

2.3. Rotation-Based Quantization

Quantization methods based on geometric transformations have attracted increasing attention in recent years [35]. QuaRot [19] is the first to systematically introduces random orthogonal rotation, projecting activation, weight, and KV cache from the original space into a more balanced feature space to alleviate dynamic range imbalance caused by extreme outliers concentration. QuIP [33] leverages structured projections to reconstruct weight bases, demonstrating favorable properties for low-bit weight quantization. SpinQuant [20] extends this line of work by introducing learnable rotation fusion, achieving strong empirical performance across multiple LLMs benchmarks. Although SpinQuant [20] additionally applies rotation to KV-cache quantization to further suppress outliers, the practical benefit of this strategy remains limited.First, KV-cache quantization primarily targets long-context inference scenarios, and its impact on memory footprint and accuracy is relatively minor for standard inference workloads. Second, under commonly adopted FP8 KV-cache quantization configurations, the quantization error itself is already negligible. Considering the trade-off between performance gains and implementation complexity, the engineering value of introducing online rotation operators for KV-cache quantization warrants further scrutiny. As a result, our method does not incorporate additional rotation specifically for KV-cache optimization. We find that applying rotation-based quantization to weight and activation alone already yields substantial stability improvements, capturing most of the performance gains of SpinQuant [20] under the target precision settings, while maintaining a simpler and more efficient implementation.

While Rot-KDEQ also leverages orthogonal rotation to mitigate activation outliers, our method is conceptually and technically distinct from prior rotation-based approaches such as QuaRot [19], SpinQuant [20], and ResQ [6]. Existing methods primarily focus on geometric transformations or mixed-precision strategies to alleviate quantization error, without explicitly modeling the underlying activation distribution. In contrast, Rot-KDEQ introduces per-channel KDE to statistically characterize the high-density core region of activations after rotation. This density-aware design enables adaptive quantization range construction and hybrid scaling under ultra-low-bit settings. As a result, Rot-KDEQ complements rotation-based isotropization with explicit distribution modeling, addressing limitations that cannot be resolved by rotation alone.

3. Method

In this section, we describe the design of Rot-KDEQ, a post-training quantization framework for ultra-low-bit inference. Rot-KDEQ consists of two components: **(i)** random orthogonal rotation applied to weights and activations, and **(ii)** per-channel construction of activation quantization ranges based on kernel density estimation (KDE).These two components operate synergistically at both the geometric and statistical levels, enabling more stable quantization performance under 4-bit precision and below. Following prior works such as QuaRot [19], we adopt orthogonal rotation as a preprocessing step, while our main contribution lies in integrating rotation with density-aware activation modeling for robust ultra-low-bit quantization.

3.1. Quantization Fundamentals

In Post-Training Quantization(PTQ) of LLMs, the primary quantization targets include model weight, activation, and key value (KV) cache. These tensors are mapped to low-bit integer

representations, substantially reducing memory footprint and computational complexity during inference. Specifically, given a real-valued matrix $\mathbf{X}$, the $\mathbf{b}$-bit integer quantization and dequantization process can be formulated as:

$$Q_b(\mathbf{X}) = s_{\mathbf{X}} \cdot (\left\lfloor \frac{\mathbf{X}}{s_{\mathbf{X}}} + z_{\mathbf{X}} \right\rceil - z_{\mathbf{X}}) \tag{1}$$

where $[\cdot]$ denotes the rounding-and-clipping operator, and $s_{\mathbf{X}}$ and $z_{\mathbf{X}}$ represent the quantization scale and zero-point, respectively.

For symmetric quantization, which is commonly adopted in LLMs [36, 37], the quantization parameters are defined as:

$$s_X = \frac{\max(|X|)}{2^{b-1} - 1}, z_X = 0. \tag{2}$$

This formulation assumes that the underlying distribution is centered around zero and approximately symmetric. In contrast, for modules that exhibit asymmetric distributions [38, 39], the quantization parameters are configured as:

$$s_X = \frac{\max(X) - \min(X)}{2^b - 1}, z_X = \text{clip}\left(\left\lfloor -\frac{\min(X)}{s_X} \right\rceil, 0, 2^b - 1\right). \tag{3}$$

which enables more accurate adaptation to shifted and skewed activation distributions.

All methods proposed in this work are developed and optimized under the asymmetric quantization framework, which provides greater flexibility for modeling the highly skewed activation distributions commonly observed in LLMs.

3.2. Random Orthogonal Rotation: Mitigating Activation Skewness and Outlier Dominance

Given an input activation tensor $X \in \mathbb{R}^{n \times d}$ and a weight matrix $W \in \mathbb{R}^{d \times d}$, we first project both tensors into a shared orthogonal basis using a random orthogonal matrix $R \in \mathbb{R}^{d \times d}$. After this transformation, the numerical distributions of activation and weight become closer to Gaussian-like and more isotropic, which significantly reduces quantization error when they are uniformly quantized to 4-bit precision, thereby effectively mitigating accuracy degradation.

Due to the orthogonality of R, the following identity holds: $RR^T = RR^{-1} = I$. The quantized activation is then given by:

$$X_q = Q_b\left(XR\right)_{KDE} \cdot R^T. \tag{4}$$

Where $(\cdot)_{KDE}$ denotes the KDE-based clipping and quantization operation, which will be detailed in the next subsection.

Similarly, the quantized weight matrix is obtained by rotating the input space of the weight and quantizing the transformed coefficients:

$$W_q = R \cdot Q_b(R^T W) \tag{5}$$

The output of the linear layer can therefore be expressed as:

$$X_q W_q = Q_b\left(XR\right)_{KDE} \cdot Q_b(R^\top W) \tag{6}$$

Notably, the quantized output in the above formulation can be decomposed into two independent components, $Q_b\left(XR\right)_{KDE}$ and $Q_b(R^T W)$, which correspond to the quantization of activation and weight in the rotated space, respectively. The key insight of this design is that all tensors are projected into the same random orthogonal space prior to quantization, ensuring that subsequent quantization is performed in a more uniform and standardized environment. This not only theoretically promotes independence of quantization errors across dimensions, but also enables efficient implementation using standard high-performance computation kernels [33].

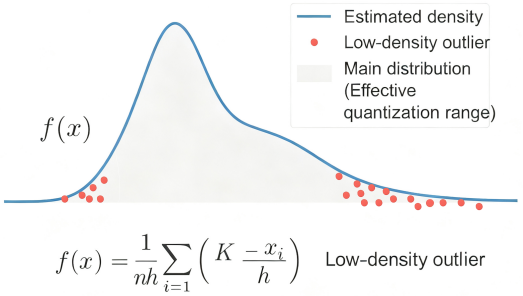


Figure 3: **Illustration of how KDE separates the main distribution region from outliers.** The blue curve denotes the KDE-estimated probability density $f(x)$, red dots represent low-density outliers, and the gray area marks the main distribution range (effective quantization interval).

3.3. KDE-Based Per-Channel Outlier Threshold Detection

Activation quantization in post-training settings requires careful handling of highly non-uniform value distributions, where quantization parameters are sensitive to extreme values.To address this problem, we introduce a KDE-based outlier detection mechanism to model and analyze activation distributions prior to quantization. Compared to conventional approaches that rely on mean and variance statistics, the key advantage of KDE lies in its non-parametric nature, which directly estimates the underlying probability density via kernel functions and thus more faithfully captures the complex distributional patterns of internal tensors in deep models.

Formally, given a set of activation samples $\{x_i\}_{i=1}^n$, we estimate the probability density function using a Gaussian kernel:

$$\begin{aligned} \hat{f}(x) &= \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right), \\ K(x) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \end{aligned} \tag{7}$$

where h denotes the bandwidth parameter. For each sample, we compute its log-density $\ell_i = \log \hat{f}(X_\rangle)$, where lower values indicate that the sample resides in a sparser region of the distribution.Based on this density metric, we identify outliers as samples whose log-density falls below a predefined quantile threshold τ, forming the outlier set $O = \{X_\rangle \mid \ell_i \leq \tau\}$. lowest percentilelowest percentile, while the remaining samples

constitute the inlier set $\mathcal{I} = \mathcal{X}/\mathcal{O}$.We then define the effective quantization range using the minimum and maximum values of the inlier set $[r_{max}, r_{min}]$. ensuring that the resulting scale is primarily governed by the high-density region rather than being stretched by a small number of extreme outliers.

This KDE-based outlier detection mechanism plays a dual role in the quantization framework. First, it enhances adaptability to the dynamic variability of activation distributions. Since activation tensors can vary substantially across layers and inputs, global clipping strategies often lack robustness, whereas KDE flexibly adjusts quantization ranges by explicitly capturing local density structure. Second, it improves the robustness of quantization parameters: by prioritizing coverage of high-density regions, the method significantly reduces quantization error for the majority of elements while still retaining a principled mechanism to handle outliers, preventing complete loss of critical information during inference.

Notably, when the inlier set I becomes extremely small or nearly empty, the method naturally degenerates to global quantization, directly adopting $[X_{max}, X_{min}]$ as the quantization range. This property ensures the generality and stability of the proposed approach. By embedding KDE-based outlier detection into the quantization process, we can substantially reduce activation quantization error without introducing additional training overhead, thereby achieving improved performance under ultra-low-bit PTQ settings.

3.4. Hybrid Quantization Factors and Piecewise Dequantization(HQFM)

To support piecewise quantization under non-uniform activation distributions, we introduce a hybrid quantization factor mechanism (HQFM). HQFM applies different quantization factors to different regions of the activation domain, enabling separate handling of the main distribution region and tail values.

Using the KDE-based procedure described in Sec. 3.3, we first identify an effective main activation interval $[r_{\min}, r_{\max}]$. Based on this interval, two sets of quantization factors are defined:

- **Local quantization factors**, derived from $[r_{\min}, r_{\max}]$, which are applied to values within the main distribution region;
- **Global quantization factors**, derived from $[X_{\min}, X_{\max}]$, which are applied to values outside the main region.

To select the appropriate factor during inference, we define a binary interval-selection mask:

$$\mathbb{M}(x) = \begin{cases} 1, & r_{\min} \le x \le r_{\max}, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

For activation values within the main region, local quantization factors are applied. For values outside this region, global quantization factors are used. The resulting piecewise quantization and dequantization process is defined as:

$$\hat{x} = \mathbb{M}(x) \cdot \tilde{x}_{\text{local}}(q_{\text{local}}(x)) + (1 - \mathbb{M}(x)) \cdot \tilde{x}_{\text{global}}(q_{\text{global}}(x)). \tag{9}$$

This formulation naturally extends to tensor-wise operations and can be efficiently implemented using vectorized computation.

Algorithm 1 Kernel Density Estimation Outlier Detection

Require: Data array **data**, bandwidth $bw = 0.2$, percentile $p = 5$

Ensure: Outlier values **outliers**, outlier indices **outlier_indices**

Convert input data to float: **data** ← float32(**data**)
Reshape to 2D: **X** ← reshape(**data**, −1, 1)
  kde ← KernelDensity('*gaussian*', = *bw*)
Fit KDE to data: **kde**.fit(**X**)
  log_densities ← **kde**.score_samples(**X**)
  τ ← percentile(**log_densities**, p)
  outlier_indices ← where(**log_densities** ≤ τ)
  outliers ← **data**[**outlier_indices**]
return outliers, **outlier_indices**

3.5. Fusion of Rotation Matrices and Efficient Implementation

In this work, we adopt a simplified variant of the QuaRot framework [19]. Given the decoder-only architecture of LLMs, we retain only two types of projection matrices:

- the inter-block projection matrix $\mathbf{R}_A \in \mathbb{R}^{d_h \times d_h}$, where d_h denotes the hidden dimension,
- the intra-FFN projection matrix $\mathbf{R}_D \in \mathbb{R}^{d_{\text{FFN}} \times d_{\text{FFN}}}$,where d_{FFN} is the intermediate dimension of the feed-forward network.

As discussed in Sec. 2.3, we do not apply rotation to the KV cache.As illustrated in Fig. 2, $\mathbf{R}_A$ is applied to the input activation of both the self-attention and feed-forward blocks, i.e., at the boundaries of Transformer blocks. To avoid introducing additional inference-time overhead for activation processing, we absorb this projection into the weight of the last linear layer of the preceding block via post-multiplication. Specifically, given an input activation $\mathbf{X}$ and weight matrix $\mathbf{W}$, we have

$$\mathbf{Y} = \mathbf{X}(\mathbf{W}\mathbf{R}) \quad \Longleftrightarrow \quad \mathbf{Y} = (\mathbf{X}\mathbf{W})\mathbf{R}, \tag{10}$$

which effectively applies the projection to the output of the previous block and, equivalently, to the input of the current block.

To preserve numerical equivalence, we introduce the inverse projection $\mathbf{R}_A^T$. By pre-multiplying the weight of the first linear layer in each block with $\mathbf{R}_A^T$, the effect of $\mathbf{R}_A$ is exactly canceled. Through this coordinated modification of adjacent linear layers, the projection operation is fully absorbed into the model weight, resulting in zero additional inference cost.

Since residual connections propagate activation across layers throughout the entire model, it is also necessary to apply corresponding projections to the embedding layer and the output layer weight to ensure consistency within the residual stream. This guarantees that the inserted projections do not disrupt the model structure or functionality, enabling accurate and efficient ultra-low-bit quantization.

Method	W4/A4/KV4 (↓)					W3/A3/KV3 (↓)				
	LLaMA-2 7B	LLaMA-2 13B	LLaMA-2 70B	LLaMA-3 8B	LLaMA-3.2 3B	LLaMA-2 7B	LLaMA-2 13B	LLaMA-2 70B	LLaMA-3 8B	LLaMA-3.2 3B
Baseline	5.5	5.0	3.3	6.10	7.80	5.5	5.0	3.3	6.10	10.70
RTN [40]	2e3	3e3	2e5	218.9	268.8	1e4	1e4	1e6	1e3	1e3
GPTQ [22]	9e3	3e3	1e6	166.3	78.3	1e5	1e4	1e7	529.8	357.9
SmoothQuant [14]	15.4	11.2	10.5	78.2	96.1	-	-	-	269.4	289.1
LLM-QAT [41]	14.9	4e3	-	52.5	42.0	30.5	24.9	22.2	187.3	86.2
OmniQuant [32]	11.2	5.4	-	-	-	18.9	15.2	13.7	-	-
QuIK [42]	7.5	6.8	6.9	14.2	15.8	14.1	12.5	10.2	25.6	28.4
QuaRot [19]	6.3	5.4	3.7	7.8	10.1	13.5	11.5	9.4	15.2	18.3
SpinQuant [20]	6.0	5.2	3.8	7.3	11.4	13.0	11.0	9.9	16.9	20.3
ResQ [6]	**5.9**	5.1	3.8	7.1	8.8	13.0	10.6	8.6	14.6	17.5
Rot-KDEQ (Ours)	**5.9**	**5.0**	**3.5**	**6.9**	**8.5**	**11.1**	**8.2**	**6.8**	**12.3**	**15.2**

Table 2: **Performance comparison of Post-Training Quantization(PTQ) methods on LLaMA models under ultra-low-bit settings.**This table compares representative PTQ methods on LLaMA-2 (7B/13B/70B), LLaMA-3, and LLaMA-3.2 under W4A4KV4 and W3A3KV3 configurations. As bit-width decreases, most methods suffer substantial performance degradation, whereas Rot-KDEQ consistently achieves the lowest or near-lowest perplexity across all model scales and settings, demonstrating strong robustness, scalability, and generalization under aggressive quantization.

As also shown in Fig. 2, $\mathbf{R}_D$ is applied to the input activation of the feed-forward network. Unlike $\mathbf{R}_A$, this projection cannot be fused into the weight of the previous block. This is because the input to the `down_proj` layer first passes through a non-linear activation function, and non-linearities break the commutativity required to absorb the projection into the preceding `up_proj` weight. Consequently, $\mathbf{R}_D$ must be applied online to the FFN input activation, i.e., computing $\mathbf{X} \cdot \mathbf{R}_D$ at inference time. To compensate for this projection, the weight of the `down_proj` layer are pre-multiplied by $\mathbf{R}_D^T$.

However, since d_{FFN} is typically very large, directly applying a dense $d_{\text{FFN}} \times d_{\text{FFN}}$ matrix would incur prohibitive computation and storage costs. To address this issue, we design $\mathbf{R}_D$ as an online Hadamard matrix. Multiplication with such a matrix can be efficiently implemented using the Fast Hadamard Transform (FHT), with a computational complexity of $O(n \log n)$, which is substantially lower than the $O(n^2)$ complexity of standard dense matrix multiplication, while completely avoiding the need to store a large dense matrix.

4. Experimental Results

4.1. Experimental Environments

We conduct a comprehensive evaluation of the proposed Rot-KDEQ method on a diverse set of mainstream LLMs, including OPT [43], BLOOM [44], LLaMA-2 [45], and the recent LLaMA-3.2 family [46]. This selection spans different model scales, architectural designs, and release stages, enabling a thorough assessment of the stability, generality, and scalability of the proposed approach.

All experiments are performed on standard language modeling benchmarks, including WikiText-2 [47], Penn Treebank (PTB) [48], and C4 [49], with perplexity (PPL) used as the sole evaluation metric. Perplexity directly reflects predictive uncertainty in generative language modeling tasks and is therefore well suited for measuring the impact of quantization error on semantic modeling capability. This unified evaluation protocol avoids confounding factors introduced by multi-task benchmarks, allowing for more direct and fair comparisons across different quantization methods.

For baselines, we include a representative set of Post-Training Quantization(PTQ) approaches for LLMs, covering a broad spectrum of design paradigms:SmoothQuant [14] , GPTQ [22], LLM-QAT [41], QuIP [33], QuaRot [19], SpinQuant [20],RTN (Round-to-Nearest) [40] and ResQ [6]. These methods encompass weight reconstruction, scale migration, rotation-based transformations, and mixed-precision strategies, collectively characterizing the current performance boundaries of PTQ methods under ultra-low-bit settings.

In terms of implementation, Rot-KDEQ is developed by extending the quantization parameter configuration of QuaRot [19] and is implemented in PyTorch [50]. All experiments are conducted on a single NVIDIA RTX A6000 GPU without model parallelism or multi-GPU acceleration, ensuring consistent experimental conditions and reproducibility of the reported results.

4.2. Performance Comparison in LLMs

Tab. 2 and Tab. 3summarize the perplexity (PPL) comparisons between Rot-KDEQ and a range of representative PTQ methods across different model families and ultra-low-bit quantization configurations. We note that the exact quantization setups vary slightly across model series, and the detailed configurations are explicitly specified in each table. In this section, we focus on analyzing how different methods degrade as quantization precision decreases under practical evaluation settings, with particular emphasis on their relative stability in preserving language modeling performance.

On the LLaMA-3 and LLaMA-3.2 model families (Tab. 2), we evaluate two representative configurations: W4A4KV4, which can be regarded as a moderately ultra-low-bit setting, and the more challenging W3A3KV3 configuration. The results show that even under W4A4KV4, noticeable performance gaps already emerge among different methods. Rot-KDEQ achieves the lowest perplexity across both model scales, indicating that channel-wise heterogeneity of activation distributions remains a critical factor limiting quantization accuracy in modern model architectures, and that approaches relying solely on Empirical scale adjustment or global geometric transformations leave room for improvement.

Method	bits(W/A/KV)	OPT (↓)				BLOOM (↓)	
		125M	1.3B	2.7B	6.7B	560M	1.1B
Full-Precision	16/16/16	27.65	14.63	12.47	10.86	22.42	17.69
GPTQ [22]	4/4/4	31.12	15.47	12.87	11.39	24.03	19.05
QuIP [33]	4/4/4	33.35	16.21	14.55	12.86	26.72	22.09
Rot-KDEQ(ours)	4/4/4	**28.48**	**14.98**	**12.69**	**11.02**	**23.71**	**18.55**
GPTQ [22]	3/3/3	53.85	15.47	12.87	11.39	32.31	25.08
QuIP [33]	3/3/3	30.92	17.12	15.63	13.30	30.26	22.27
Rot-KDEQ(ours)	3/3/3	**29.87**	**15.04**	**12.84**	**11.28**	**26.04**	**21.61**

Table 3: **Comparison of perplexity (PPL) for different Post-Training Quantization(PTQ) methods on the OPT and BLOOM model families.**Across multiple model scales, this table compares GPTQ, QuIP, and Rot-KDEQ under W4A4KV4 and W3A3KV3 ultra-low-bit configurations. The results show that Rot-KDEQ consistently outperforms existing methods under 4-bit settings, and its performance advantage becomes even more pronounced as the compression level is further increased to 3-bit, demonstrating the superior robustness and stability of the proposed rotation and density-aware quantization mechanism under ultra-low-bit conditions.

When the quantization precision is further reduced to W3A3 KV3, these performance gaps are significantly amplified. Several baseline methods exhibit sharp increases in perplexity, with some approaching or reaching unusable regimes. This behavior reflects the fact that under ultra-low-bit conditions, the dominance of a small number of extreme activation outliers over quantization ranges becomes difficult to suppress using conventional scale design alone. In contrast, Rot-KDEQ maintains stable and substantially lower perplexity under this setting, demonstrating that the density-aware quantization ranges constructed by combining random orthogonal rotation with per-channel Kernel Density Estimation(KDE) effectively mitigate error amplification caused by extreme outliers.

Similar trends are consistently observed on the OPT and BLOOM model families (Tab. 3). As model size increases or quantization bit-width decreases, performance degradation of existing methods generally becomes more severe, whereas the relative advantage of Rot-KDEQ correspondingly widens. These results suggest that the proposed approach is not confined to specific models or configurations, but instead provides more reliable control over quantization-induced performance loss in scenarios where activation distributions are more complex and quantization errors are more prone to accumulation.

Based on the results in Tab. 2 and Tab. 3, we summarize three key observations:

1. The relative advantage of Rot-KDEQ increases as quantization precision decreases. Under moderately ultra-low-bit settings, performance differences among methods remain limited; however, in ultra-low-bit configurations such as W3A3KV3, most baseline approaches suffer severe or even catastrophic degradation, while Rot-KDEQ continues to deliver stable and acceptable perplexity.
2. Rot-KDEQ exhibits consistent stability across different model families and architectural variants. Whether evaluated on legacy architectures such as OPT and BLOOM, or on more recent models including LLaMA-2, LLaMA-3, and LLaMA-3.2, Rot-KDEQ consistently achieves the lowest or near-lowest perplexity under corresponding evaluation setups, indicating strong cross-model generalization without reliance on architecture-specific assumptions.
3. Quantization strategies based solely on global geometric balancing remain insufficient under ultra-low-bit settings. In contrast, Rot-KDEQ jointly addresses activation distribution skewness and heterogeneity at both global and local scales by combining random orthogonal rotation with per-channel density-aware quantization ranges. This joint design is a key factor enabling its stable performance under complex distributional conditions.

4.3. Ablation Study

To further analyze the contributions of individual components in Rot-KDEQ, we conduct an ablation study under the LLaMA-2 7B model with the W4A4 quantization configuration. We compare quantization performance across different module combinations with the following ablation settings:

- Baseline: Conventional layer-wise extremum-based quantization without rotation or KDE modeling.
- Rotation-Only: Incorporating random orthogonal rotation to achieve global distribution balancing.
- KDE-Only: Applying per-channel KDE for adaptive quantization range construction without rotation.
- Full Rot-KDEQ: Enabling random orthogonal rotation, per-channel KDE, and the hybrid quantization factor mechanism.

Method Variant	Rotation	KDE	PPL(↓)
Baseline (MinMax)	×	×	1.7e3
Rotation Only	✓	×	8.87
KDE Only	×	✓	7.15
Rot-KDEQ (Full)	✓	✓	**5.90**

Table 4: Ablation study of different components in Rot-KDEQ evaluated on LLaMA-2 7B under the W4A4KV4 setting.

The results show that introducing rotation alone leads to a moderate reduction in perplexity, indicating that global energy dispersion helps alleviate scale mismatch caused by concentrated outliers. In contrast, the KDE-only variant yields more pronounced performance gains, further validating the effectiveness of per-channel statistical modeling in capturing the true activation distribution and tightening quantization ranges.

When both components are combined to form the complete Rot-KDEQ framework, the model achieves the lowest perplexity. These results clearly demonstrate the strong complementarity between global rotation-based balancing and local density-adaptive modeling, and confirm that their joint design is a key factor in achieving stable and effective quantization under ultra-low-bit settings.

4.4. Hyperparameter Analysis

To evaluate the sensitivity and stability of Rot-KDEQ with respect to key hyperparameters, we further conduct a systematic analysis of the bandwidth parameter used in KDE.

The experimental results show that when the KDE bandwidth is varied within a reasonable range, model perplexity changes only marginally, with the performance curve exhibiting a clear plateau region. This behavior indicates that Rot-KDEQ is robust to bandwidth selection and does not require fine-grained hyperparameter tuning to achieve stable performance.

Model	σ				
	0.05	**0.1**	**0.2**	**0.3**	**0.5**
LLaMA2-7B(↓)	6.5	6.1	**5.9**	6.7	6.7
OPT-125m(↓)	28.9	28.5	**28.4**	**28.4**	29.3

Table 5: Sensitivity analysis of KDE bandwidth parameter evaluated on LLaMA-7B under W4A4KV4 setting.

These results demonstrate that Rot-KDEQ exhibits strong insensitivity to hyperparameter variations, without suffering from performance instability caused by parameter fluctuations. This property further enhances the practical deployability of the proposed method, allowing it to adapt to LLMs of different scales without complex manual tuning.

5. Discussion

5.1. Limitations of Purely Geometric Quantization under Ultra-Low-Bit Settings

Rotation-based quantization methods have demonstrated strong effectiveness in mitigating extreme activation outliers by redistributing feature energy across dimensions. By projecting activations into a more isotropic space, these approaches substantially reduce the dominance of individual channels and improve robustness under moderate ultra-low-bit settings. However, under ultra-low-bit constraints, such as 4-bit precision and below, geometric reorganization alone becomes insufficient.

The primary limitation lies in the fact that rotation operates at a global geometric level and does not explicitly account for the statistical density of activation values within each channel. After rotation, quantization range selection is still typically governed by global extrema or empirical thresholds. When bit-width is severely constrained, even minor misalignment between quantization intervals and the high-density activation mass can lead to disproportionate precision loss. This observation explains why rotation-based methods may still experience instability under aggressive compression despite achieving improved isotropy.

5.2. Importance of Density-Aware Quantization Range Design

Beyond geometric imbalance, ultra-low-bit quantization fundamentally requires precise alignment between quantization ranges and the dominant activation distribution. Empirical strategies such as clipping range selection implicitly assume simple distribution shapes and fail to adapt to the complex, non-Gaussian, and often multi-modal activation patterns observed in large language models.

Explicitly modeling activation density provides a principled mechanism for distinguishing high-density core regions from low-density tails. By anchoring quantization intervals to the statistically dominant regions rather than extreme values, density-aware range design enables more efficient allocation of limited quantization levels. This perspective highlights that robust ultra-low-bit quantization is not solely a geometric problem, but also a statistical one that requires explicit consideration of local distribution structure.

5.3. Complementarity between Geometric Rebalancing and Statistical Modeling

The above analysis suggests that geometric rebalancing and statistical modeling address two orthogonal sources of quantization error. Geometric transformations primarily mitigate global energy concentration and channel dominance, while density-aware modeling governs how quantization resources are allocated within each channel. Neither component alone is sufficient to guarantee stable ultra-low-bit performance.

By decoupling these two factors, quantization frameworks can more effectively control error propagation under severe precision constraints. Geometric reorganization establishes a balanced representation space, within which statistical modeling can accurately identify and preserve the dominant activation mass. This complementarity provides a unified perspective on why joint consideration of geometric and statistical properties is essential for reliable post-training quantization in ultra-low-bit regimes.

5.4. Limitations and Future Directions

While the proposed framework demonstrates strong robustness under aggressive quantization, several limitations remain. First, density estimation introduces additional computational overhead during calibration, which may become non-negligible for extremely large models or limited calibration budgets. Second, although the current design effectively handles static activation distributions, extending density-aware modeling to dynamic or input-dependent distributions remains an open challenge.

Future work may explore lightweight or approximate density modeling techniques, as well as adaptive strategies that integrate distribution modeling with online inference constraints. Investigating the interaction between density-aware quantization and emerging hardware primitives also represents a promising direction for further improving deployment efficiency.

6. Conclusion

In this work, we presented Rot-KDEQ, a unified post-training quantization framework designed for reliable ultra-low-bit inference of large language models. By jointly addressing global geometric imbalance and local statistical heterogeneity, Rot-KDEQ enables stable quantization under aggressive precision constraints without requiring retraining.Our analysis highlights that effective ultra-low-bit quantization requires decoupling geometric reorganization from quantization range selection, and that explicitly modeling activation density is critical for preserving the dominant activation mass. This perspective provides a principled explanation for the limitations of existing PTQ approaches under extreme compression.We believe that Rot-KDEQ offers a practical and general solution for ultra-low-bit LLMs deployment, and that the underlying design principles may inspire future research on distribution-aware quantization and efficient inference on resource-constrained hardware.

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